05. Validation Strategy for Unsupervised Learning

Why there is no conventional label-based train/test split—and how to validate segmentation honestly.

Project: SmartCart Customer Segmentation

Source checked: Sai-Likhith-2005/Customer-Segmentation | smartcart.ipynb | smartcart_customers.csv

What the repository does

The notebook does not create a train/test split or a cross-validation loop. It fits median imputation, feature engineering, one-hot encoding, StandardScaler and PCA using all 2,240 customer records (then filters to 2,236), and clusters those same records.

Why ordinary supervised splitting is not the default

There is no ground-truth segment label to predict. The algorithm creates groups from the feature geometry. Therefore, classification accuracy against a target label is not defined. Validation instead asks whether clusters are compact, separated, stable, interpretable and useful for a decision.

A defensible validation plan

- Create a reference cohort and a holdout cohort by customer or time, depending on intended use.
- Fit imputation, encoding, scaling and PCA on the reference cohort only.
- Choose K and fit the clustering algorithm on the reference cohort; do not use the holdout to tune repeatedly.
- Assign holdout customers using a method supported by the clustering algorithm. K-Means can assign to nearest centroids; AgglomerativeClustering does not provide a native predict method, so use an explicit assignment strategy or choose a predictive surrogate.
- Compare holdout cluster profiles and distances to centroids; flag customers far from every cluster.
- Repeat the analysis across bootstrap samples or time windows to test stability.

Metrics and diagnostics

Diagnostic	What it tells you	Caution
Silhouette	Within-cluster cohesion vs nearest-cluster separation	Can favour simple shapes; not business value
Davies–Bouldin	Average similarity of each cluster to its most similar cluster	Lower is better; depends on distance geometry
Calinski–Harabasz	Between-cluster dispersion relative to within-cluster dispersion	Higher is better; compare on same representation
Stability / ARI	Agreement of assignments across resamples	Cluster IDs may permute; align labels before comparison
Business profiling	Whether groups differ in useful behaviours	Differences may be small or not actionable

Project-specific limitation

The notebook plots silhouette scores for K=2 through 10, but the saved output does not print the numeric scores. The only printed selection result is KneeLocator's K=4. It is therefore incorrect to claim that the silhouette score independently selected K=4.

Interview explanation

“Because this is unsupervised learning, I did not have a true segment label for a conventional train/test accuracy calculation. I would validate cluster quality with silhouette and other internal metrics, test stability across samples and time, and check whether the resulting profiles support distinct business actions.”